

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

Anonymous authors
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ABSTRACT

Parametric imitation learning via behavior cloning (BC) is fundamentally constrained by compounding error: because the policy is supervised only at expert states, closed-loop deployment inevitably drives the agent into out-of-distribution regions in which its predictions destabilize and its generalization deteriorates. We show that this limitation can be resolved without relaxing any assumption of standard BC—by reusing the demonstration data at inference time through a semi-parametric, retrieval-based formulation. We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, which recasts imitation learning around the local neighborhood structure of the demonstrations rather than a single global state-to-action mapping. Instead of compressing the demonstrations into one parametric function and discarding them at test time, DARP predicts each action from the k -nearest neighbors of the query state within the expert dataset, conditioning every neighbor-level prediction on the neighbor state, its associated expert action, and the relative difference vector between neighbor and query, and then combining these neighbor-informed predictions through a permutation-invariant aggregator. This construction simultaneously grounds every prediction in observed data and implicitly enforces local consistency, and it does so strictly within the BC regime: it requires no additional data collection, no online expert feedback, and no task-specific knowledge. We further establish that this neighbor aggregation implements a parameter-free Laplacian filter on the k -nearest-neighbor graph of expert states, which explains precisely why it suppresses variance and stabilizes rollouts. Across continuous control and robotic manipulation—spanning both simulation and real hardware, and representations that range from compact low-dimensional states to high-dimensional visual features—DARP delivers consistent improvements of 15–46% over standard behavior cloning, with its largest relative gains arising in exactly the high-dimensional visual regime where BC is most brittle. Code and demos will be released upon acceptance.

1 INTRODUCTION

Imitation learning via behavior cloning (BC) (Pomerleau, 1991) has enabled robots to acquire complex, dexterous behaviors directly from expert demonstrations (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014). Yet for all its appeal, BC remains brittle at deployment, and markedly so on long-horizon tasks (Ross et al., 2011). The mechanism responsible is *covariate shift*: because the policy is supervised only at expert states, the small prediction errors incurred during a rollout compound, progressively driving the agent into regions that the demonstration data represents poorly (Spencer et al., 2021; Ross et al., 2011). In precisely these out-of-distribution regions the policy destabilizes, emitting high-variance actions that repeatedly precipitate task failure. This paper shows that the failure is not intrinsic to the BC objective but to how the demonstrations are used, and that it can be corrected without stepping outside the BC regime at all.

Although this failure mode is well recognized, the prevailing remedies for compounding error (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020) step outside the standard BC assumptions, presupposing access to simulators, interactive experts, large quantities of sub-optimal data, or strong task-specific structure. We adopt a deliberately more demanding stance: to remain squarely within the pure BC regime, learning only from expert state–action pairs with no additional supervision or feedback. This commitment sharpens the central question of the paper: *can*

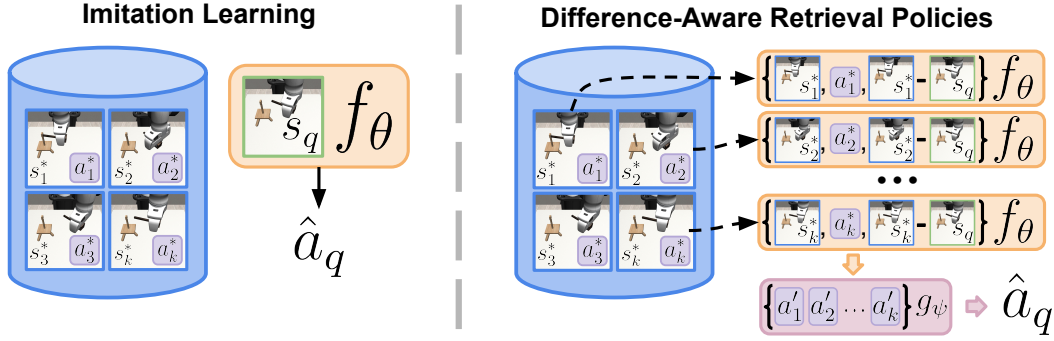


Figure 1: **Overview of DARP.** Standard BC (left) compresses the expert state–action dataset into a single parametric function during training and discards the data at inference time. DARP (right) instead retains the dataset and, at each query point s_q , retrieves a local neighborhood of expert state–action pairs, reparameterizing the prediction around the difference vectors between the query state and its retrieved neighbors. It is precisely this grounding of predictions in retrieved data that keeps DARP stable in the out-of-distribution regions where global BC policies degrade.

the variance of a BC policy be reduced using nothing beyond the original demonstration dataset? We answer this question affirmatively.

The statistical origin of the problem points directly to the answer. Behavior cloning minimizes only the supervised risk on expert states; this controls bias on the training distribution but leaves variance entirely unchecked, so in the low-density regions routinely visited during closed-loop rollouts the learned policy is free to oscillate arbitrarily. The principled corrective is to impose *smoothness*, mapping nearby states to similar actions (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024) so as to suppress these spurious fluctuations and stabilize rollouts. A range of mechanisms for encouraging such smoothness has been explored, which we survey in Section 4.

Each such mechanism, though effective within its own regime, carries a characteristic limitation: data augmentation cannot guarantee consistency, global priors tend to blur behaviorally distinct modes, temporal penalties regularize only along the time axis rather than across the state space, and explicit graph regularizers introduce additional smoothness hyperparameters that must be tuned. These limitations motivate a design that secures smoothness without any of these costs.

A complementary axis in the literature contrasts global with local learning. “Global” supervised models (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) compress the entire demonstration dataset into a single parametric function and are, for exactly that reason, brittle under distribution shift. “Local” methods (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994) instead adapt each prediction to the structure of the dataset, consulting neighborhoods of similar states and forming outputs through non-parametric or semi-parametric operations on the empirical distribution of expert behavior. Locality confers robustness precisely because it dispenses with reliance on a single parametric function; the local approach, however, carries its own well-defined limitations: its accuracy is tied to the choice of distance metric, naive neighborhood averaging blurs distinct actions and cannot represent multimodality, and characterizing neighbors solely by their absolute states constrains generalization. The formulation we introduce is designed to inherit the robustness of locality while resolving each of these limitations.

We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a semi-parametric formulation that unifies the robustness of local methods with the stability of regularized global policy learning. Rather than mapping the current query state to an action through a single feedforward pass, DARP (Fig. 1) first retrieves a set of k neighbors from the training corpus and then produces k candidate actions, each conditioned on a tuple comprising the neighbor state, its associated expert action, and its difference from the query state; a permutation-invariant aggregator then combines these candidates into a single robust prediction. This design achieves two objectives at once: non-parametric retrieval grounds every prediction in observed data, while parametric action prediction conditioned on the retrieved neighbors enforces local consistency. Decisively, it does so while remaining entirely within the assumptions of standard behavior cloning—no additional data, no online supervision, and

no task-specific knowledge are required. In spectral terms, this form of neighbor aggregation realizes a Laplacian filter on the k -nearest-neighbor (k -NN) graph of expert states, yielding a parameter-free smoother that adapts automatically to the local density and geometry of the dataset.

We substantiate these claims with converging theoretical and empirical evidence. Operating under exactly the requirements of behavior cloning, DARP substantially improves performance by reducing variance and strengthening robustness to distribution shift. Our analysis formalizes the connection to Laplacian regularization, establishing that DARP implicitly applies a fixed low-pass spectral filter that suppresses high-frequency variance with no hyperparameter to tune. Empirically, DARP attains 15–46% gains over standard behavior cloning across continuous control (MuJoCo), robotic manipulation (Robosuite, RoboCasa, and a real-world FurnitureBench task), and high-dimensional visual imitation (Robosuite with image state), with the largest relative gains arising precisely in the high-dimensional visual regime—the setting in which BC is most brittle. Taken together, these results establish DARP as a general and scalable architecture that extends naturally to image-based domains and accommodates rich policy classes such as transformers, Gaussian mixture models, and diffusion policies. A systematic set of ablations then isolates the contributions of the representation and the architecture, yielding transferable design principles for retrieval-based algorithms in sequential decision-making. In summary, this work contributes: (i) DARP, a semi-parametric, difference-aware retrieval policy that reduces BC variance without leaving the BC regime; (ii) a spectral analysis proving that its neighbor aggregation is parameter-free Laplacian smoothing on the k -NN graph; and (iii) a comprehensive empirical study, across domains, state representations, action classes, and real hardware, that demonstrates consistent gains and identifies the components responsible for them.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

In this section we develop a new class of imitation learning methods that unifies the complementary strengths of “global” parametric learning and “local” non-parametric learning. The construction contributes a new policy architecture together with a simple training objective that learns under exactly the requirements of standard behavior cloning while delivering substantial theoretical and empirical gains, and we derive the accompanying guarantees from first principles. Readers primarily interested in the empirical results may proceed directly to Section 2.5 for a self-contained summary of DARP and then to Section 3 for the evaluation.

The development proceeds in three stages that build on one another. We first fix the problem setting (Section 2.1) and then introduce a regularized variant of imitation learning (Section 2.2) that injects structure drawn from the data, provably improving variance, generalization, and stability. Section 2.3 then establishes that these benefits can be obtained *implicitly*, by modifying the policy *architecture* rather than the objective—thereby eliminating the regularization hyperparameter altogether. Finally, Section 2.4 instantiates the practical algorithm DARP, which realizes these benefits through a semi-parametric retrieval-augmented architecture directly applicable to imitation learning with modern neural networks and generative modeling tools.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We adopt the standard imitation learning setting, formalized as a finite-horizon Markov Decision Process (MDP) $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, and P_0 the initial state distribution. A *policy* $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$ maps each state to a distribution over actions so as to maximize a task-relevant objective; for brevity we suppress the subscript θ when referring to the general function class f . We are given a dataset of expert demonstrations $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ collected as state–action pairs, and we reserve the notation s^* and a^* for the states and actions belonging to this expert dataset. Behavior cloning (Pomerleau, 1991) recovers a policy f_θ by casting imitation as supervised learning, $\arg \max_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. Although the distributional class of f_θ may be an arbitrarily expressive generative model (Lipman et al., 2023; Chi et al., 2024), we begin with a Gaussian parameterization for expository clarity¹.

¹We show that this assumption can be relaxed in Section 2.4

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Because behavior cloning minimizes the supervised imitation loss only over expert states drawn from $\mathcal{D}^*$, it exerts no control over the policy’s behavior on states that depart from the expert manifold. In practice, accumulating error drives the agent into out-of-distribution regions in which a BC policy may act arbitrarily—a pathology that is especially pronounced for overparameterized neural networks (Ross & Bagnell, 2010).

The root cause is structural: behavior cloning constrains function evaluations only at the training states and never accounts for the relationship among states, and their corresponding actions, within a neighborhood, thereby ignoring the geometry of the underlying data manifold. To inject this structure into policy learning, we augment the objective with a regularization term that explicitly enforces *local consistency*—states that are nearby in the expert dataset are mapped to similar actions. This yields the neighborhood-regularized loss $\mathcal{L}_{\text{MRIL}}$, in which the standard imitation objective $\mathcal{L}_{\text{BC}}$ is combined with a smoothness penalty $\mathcal{L}_S$ that compels predictions to respect the geometry of the dataset rather than relying on pointwise supervision alone.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \lambda \underbrace{\mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_S)}, \quad (1)$$

where $\ell(f(s^*), a^*)$ is the supervised imitation loss, $\mathcal{N}_k(s^*)$ denotes the k -nearest neighbors of s^* in the expert dataset, and the weights $w_i(s^*)$ are normalized kernel weights determined by the state differences, $w_i(s^*) \propto K_\Delta\left(\frac{\|s_i^* - s^*\|}{h}\right)$. As we establish below and develop in full in Appendix A.1.1, this penalty is a form of *manifold regularization*, or Laplacian smoothing, that penalizes high-frequency variation of the policy across the neighborhood of expert states, and it provably improves policy variance, stability, and generalization.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.
- (iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s^{*'})\| = O(\|s^* - s^{*'}\|)$ for neighbors $s^*, s^{*'}$.

² L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

Taken together, claims (i)–(iii) establish that MRIL enjoys strictly stronger generalization and stability guarantees than vanilla BC.

Proof sketch. The detailed proof is deferred to Appendix A.1.1; we sketch the argument here. The key step is to establish that the smoothness regularizer coincides exactly with a graph Laplacian penalty on the graph whose k -nearest-neighbor (k -NN) affinity matrix is defined by the kernel w_i . As the number of samples tends to infinity, this graph Laplacian penalty converges to the weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Minimizing this Dirichlet energy has two consequences: (1) the learned f is locally Lipschitz almost everywhere, which secures smoothness and hence policy stability; and (2) the penalty acts as Tikhonov regularization, reducing estimator variance while keeping bias controlled. These two consequences are exactly claims (i)–(iii). $\square$

Intuitively, the smoothness regularizer does considerably more than penalize pairwise disagreements between neighbors: it drives the learned policy toward smoothness with respect to the underlying data manifold. Concretely, it contracts the local Lipschitz constant of f along directions of high data density $p(s)$, so that small changes in state induce small, consistent changes in the predicted action. As a direct consequence, the policy generalizes more reliably on in-distribution (ID) states and extrapolates in a structured, controlled manner to the nearby out-of-distribution (OOD) states that closed-loop rollouts inevitably visit.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

Although the MRIL objective amortizes local learning and thereby improves over vanilla BC, it carries two notable drawbacks. First, it introduces a hyperparameter λ that must be tuned to balance supervised accuracy against smoothness. Second, optimizing a modified, regularized objective in place of the standard BC objective can reshape the optimization landscape in adverse ways. These drawbacks motivate a natural question: *can the benefits conferred by MRIL (Eq. 1) be obtained by modifying the policy architecture rather than the objective?*

We answer this question affirmatively. We introduce a retrieval-based modification of the policy architecture that induces an *implicit* manifold-regularization effect (iMRIL) while training under a standard imitation objective. As Fig. 2 shows, iMRIL recovers the Laplacian-smoothing benefits of MRIL using an ordinary BC loss, and does so without introducing λ as a training hyperparameter at all. We then build on this construction to develop the practical instantiation DARP in Section 2.4.

iMRIL architecture. The guiding principle of iMRIL is to relocate the neighborhood aggregation (averaging) operation from the objective, where it appears in Eq. 1, into the architecture itself. Rather than learning a standard feed-forward predictor $f(s)$ against a neighborhood-regularized smoothness objective (Eq. 1), we embed the structure of neighborhood aggregation directly into the parameterization of the action predictor $\hat{f}$ and retain the ordinary imitation learning objective. Formally, iMRIL learns the parameters of a per-state predictor f_θ such that the aggregated action predictor $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$, formed over the nearest-neighbor states $\{s_i^*\}_{i \in \mathcal{N}_k(s^*)}$ retrieved from the training set, accurately reconstructs the corresponding expert action a^* . Under this parameterization, iMRIL optimizes the following objective at training time:

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

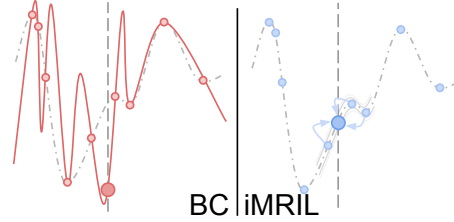


Figure 2: **iMRIL implicitly achieves Laplacian smoothing, which reduces variance and enforces local consistency**, whereas the absence of any smoothness constraint on standard BC permits arbitrarily jagged function approximations.

At deployment time, inference on a new state s_q reduces to retrieving the k -NN of s_q from the training set and applying neighborhood aggregation, $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*)$. As Section 2.4

demonstrates, the precise parameterization of f is decisive for the empirical performance of iMRIL, and it is exactly this consideration that motivates DARP.

Intuitively, the construction parameterizes the action predictor $\hat{f}$ as an aggregation of the training-neighbor predictions $f(s_i^*)$ and then learns f . Because supervision is applied to the post-aggregation function, no individual f prediction can be arbitrarily non-smooth—which is precisely what confers the benefits established in Section 2.2. We now formalize this intuition as a direct equivalence between iMRIL and the Laplacian regularization of Section 2.2.

Equivalence between iMRIL and MRIL. We defer the full proof of formal equivalence between MRIL and iMRIL to Appendix Section A.1 and state the main result, together with a proof sketch, here.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof sketch. Full details are deferred to Appendix Section A.1.2. Diagonalizing the k -NN Laplacian gives the explicit regularizer a spectral solution, yielding a filter of the form $(1 + \lambda \mu)^{-1}$ on each eigenmode. The implicit objective, in turn, can be written as neighbor aggregation $\hat{f} = Sf$ with $S = D^{-1}A$ the random-walk matrix, which shares the same eigenvectors and applies the fixed filter $1 - \mu$. Both therefore act as low-pass filters on the graph: modes with small eigenvalues, corresponding to smooth variation across the data manifold, are largely preserved, whereas modes with large eigenvalues, corresponding to rapid high-variance fluctuations between neighbors, are strongly damped. iMRIL consequently performs Laplacian smoothing implicitly, reducing variance and enforcing local consistency with no need to tune λ . $\square$

The implicit Laplacian-smoothing view does not obviate the need to learn a policy; rather, it constrains the class of functions representable after aggregation. The neighbor-conditioned network f_θ learns how expert actions vary under local perturbations, proposing a locally adapted action for each neighbor, while the aggregation operator smooths these proposals across the neighborhood to suppress variance. This induces a clean division of labor that is central to the method: learning supplies accuracy by correcting local bias, and aggregation supplies stability by controlling variance.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF iMRIL FOR IMITATION LEARNING

Building on the conceptual framework of iMRIL, we now instantiate a practical algorithm for large-scale imitation learning. Starting from the objective in Eq. 2, we make two design choices—(1) the parameterization of the per-neighbor predictor f_θ and (2) the neighbor-aggregation operator—that together drive strong empirical performance.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

The objective in Eq. 2 leaves the parameterization and input representation of f_θ unspecified. Our central observation is that neighborhood aggregation should learn how expert actions vary under local perturbations, and this in turn dictates that f_θ should exploit the *difference* between a query state

and each neighbor state in order to propose a locally adapted action per neighbor. Accordingly, in **Difference-Aware Retrieval Policies (DARP)**, rather than parameterizing f_θ solely by $f_\theta(s_i^*)$, we condition it additionally on the neighbor’s expert action a_i^* and on the difference vector $\Delta s_i = s_i^* - s_q$ between the query and neighbor states: for a query state s_q and a neighbor (s_i^*, a_i^*) , the predictor emits the action candidate $a'_i = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$. This difference-based conditioning is the design choice that, as our ablations in Section 3 confirm, most decisively determines DARP’s performance.

Let $\mathcal{N}_k(s_q)$ denote the index set of the k -nearest neighbors retrieved under a distance function $d(s_q, s_i^*)$.³ To generate predictions with DARP, we apply the neighborhood aggregation of Section 2.3 to produce an action for any query state s_q :

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

At training time, this can be used to define a straightforward imitation learning objective from the expert dataset $\mathcal{D}^*$:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2] \quad (5)$$

which optimizes the parameters of the predictor f_θ by minimizing the discrepancy between the predicted action $\hat{a}_q$ and the expert action a_q^* . Because the objective is a standard regression loss, f_θ admits any parameterization; we employ standard feedforward or convolutional networks. As Section 3 demonstrates, this difference-based parameterization is the single most important determinant of DARP’s performance. At inference time, actions are generated by retrieving the k -NN of the query state and evaluating the neighborhood aggregation of Eq. 3.

2.4.2 GOING BEYOND LINEAR AGGREGATION

Thus far, neighborhood aggregation has been restricted to a simple average of the neighbor predictions, $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$. This is a special case of a broader family of permutation-invariant aggregation functions $g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$, which may be parameterized by expressive set-equivariant models such as the Set Transformer (Lee et al., 2019) or DeepSets (Zaheer et al., 2017). The prediction model of Eq. 3 accordingly generalizes to $\hat{a}_q = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. Beyond its added expressivity, replacing the fixed average with a parametric aggregator g_ψ endows DARP with the capacity to represent richer action distributions—such as Gaussian mixture models (Pignat & Calinon, 2019) or diffusion models (Chi et al., 2024)—in place of the unimodal Gaussian implied by the L_2 -regression objective of Eq. 5. In this regime, DARP predicts not $\hat{a}_q$ directly but the parameters α of an action distribution $p(a_q; \alpha)$ —for example the means, covariances, and mixture weights of a Gaussian mixture model, or the score function of a diffusion model—and is trained by maximum likelihood over multimodal action distributions rather than by unimodal L_2 regression:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_\theta(s_q^*))], \text{ where } \alpha_\theta(s_q^*) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

Inference for a query state s_q then proceeds by sampling $\hat{a}_q \sim p(\cdot; \alpha_\theta(s_q))$, where $\alpha_\theta(s_q) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$ is constructed from the neighbors retrieved at test time. Detailed training pseudocode is provided in Appendix Section A.3.

2.5 SUMMARY

We have established that enforcing local consistency when fitting a behavior cloning policy is equivalent to manifold regularization (Laplacian smoothing) and provably improves variance, smoothness,

³We use Euclidean distance in a pre-trained embedding space, though other neighborhood functions are equally applicable; Appendix Section A.2.1 discusses the design decisions governing neighborhood construction via retrieval in detail.

and stability over standard BC (Theorem 1). Moreover, this smoothing need not be imposed through an explicit regularization term and its attendant hyperparameter λ : it can be induced implicitly by building neighborhood aggregation into the policy architecture (Theorem 2). Together these results yield the following algorithm. At training time, for each expert state–action pair (s_q^*, a_q^*) , DARP retrieves the k nearest-neighbor states from $\mathcal{D}^*$, computes the difference vector $\Delta s_i = s_i^* - s_q^*$ from each neighbor to the query, passes each neighbor tuple $(s_i^*, a_i^*, \Delta s_i)$ through the network f_θ to obtain candidate actions, and aggregates these candidates via a permutation-invariant function g_ψ into the final prediction—all trained with the standard imitation learning objective alone, without additional supervision, online feedback, or a smoothness hyperparameter.

3 EXPERIMENTAL EVALUATION

We now evaluate DARP against three questions that jointly probe its effectiveness, generality, and mechanism: **Q1**: Does DARP consistently outperform standard behavior cloning? **Q2**: Does it scale to more complex state representations and action distributions? **Q3**: Which architectural components are responsible for its performance gains? To answer them, our evaluation spans a deliberately broad range of domains, state representations, and action classes: low-dimensional states, high-dimensional image features, and both unimodal and multimodal action representations, across continuous control (MuJoCo), robotic manipulation in simulation and on real hardware (Robosuite, RoboCasa, and a real-world FurnitureBench task), and purpose-built discontinuous environments that deliberately stress-test the neighbor-based approach. This breadth lets us assess not only whether DARP helps, but where and why.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) tasks require controlling legged bodies of several distinct embodiments to achieve forward locomotion on a flat plane: Hopper (a single-legged hopping robot), Walker (a bipedal humanoid), Ant (a quadruped), and HalfCheetah (a biped). Together they span embodiments from one to four legs, testing the method across markedly different dynamics.

Robosuite Tasks: The Robosuite (Zhu et al., 2020) tasks each require a single robotic arm to manipulate objects with increasing precision. Stack requires placing a smaller cube atop a larger one; Threading requires manipulating a thin, needle-like tool and inserting it into a small ring; and Square Peg requires grasping a square wooden block with a central hole and seating it onto a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) tasks each require a single robotic arm to manipulate objects within a randomized kitchen setting, testing robustness to scene variation. Drawer requires closing an open drawer; Door requires closing an open microwave oven door; and Stove requires twisting a knob to turn off a burner.

Baseline Comparisons: We benchmark DARP against a comprehensive set of baselines and ablations that together span the non-parametric, parametric, and in-context regimes: (1) **R&P** (Sridhar et al., 2025), which directly executes the action of the nearest neighbor; (2) **LWR** (Pari et al., 2022), which performs locally weighted regression over the retrieved neighbors; (3) **BC**, standard parametric behavior cloning; (4) **REGENT** (Sridhar et al., 2025), a transformer-based in-context learning method conditioned on retrieved neighbors; and (5) **MRIL**, the explicitly smoothed variant of DARP introduced in Section 2.2. This selection isolates each axis of DARP’s design—retrieval, parametric learning, in-context conditioning, and explicit versus implicit smoothing—so that its gains can be attributed to the specific mechanism responsible.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

We first test DARP’s core hypothesis on tasks with low-dimensional state representations, a regime in which inter-state distances are well-defined and readily interpretable. The evaluation spans locomotion in MuJoCo and robotic manipulation in Robosuite and RoboCasa, with demonstrations generated via MimicGen (Mandlekar et al., 2023). Throughout these experiments the aggregation function g is deliberately restricted to a simple average of the neighbor action predictions a' , which isolates the contribution of the difference-based parameterization from that of a more expressive aggregator and thus attributes any gains to the reparameterization alone.

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: **DARP attains the highest score in every MuJoCo domain, and its Set Transformer variant is likewise the strongest or near-strongest method throughout.** DARP is compared against BC and against the non-parametric (R&P, LWR), in-context (REGENT), and explicitly smoothed (MRIL) baselines on MuJoCo continuous control using low-dimensional state. Scores are episodic returns averaged over 100 independent trials with 95% confidence intervals; higher is better, and the best result per column is in bold. Parametric policies use Multi-Layer Perceptrons; for results with diffusion policies, see Section 8. Across all four embodiments, DARP surpasses BC and every retrieval and in-context baseline, while the purely non-parametric methods (R&P, LWR) fall well behind—evidence that the difference-based parameterization, rather than retrieval alone, is responsible for the gains.

DARP improves substantially over standard behavior cloning in every environment tested, and the pattern of results answers Q1 decisively in the affirmative. The gains reach 15–25 percentage points on robotic manipulation and are especially pronounced on locomotion, where DARP attains the top score across all four MuJoCo domains (Table 1 and Table 2). The comparison also localizes the source of the advantage: the purely non-parametric methods (R&P and LWR) perform poorly throughout, confirming that naive retrieval alone is insufficient, and although MRIL almost always exceeds vanilla BC, the highest score on every task is achieved by the full DARP architecture. The difference-based parameterization, and not smoothing per se, is therefore what drives the strongest results—a conclusion the ablations of Section 3 confirm directly.

Having established these gains in the low-dimensional regime, and building on the practical instantiation introduced in Section 2.4, we next ask whether DARP scales to the higher-dimensional input representations, such as images, on which BC is most brittle.



Figure 3: The “DrawerClose” RoboCasa task: the robotic arm must close the open drawer within a randomized kitchen scene, testing manipulation under scene variation.



Figure 4: The “Threading” Robosuite task: the robotic arm must insert the needle-like implement into the small hole—a high-precision, contact-rich manipulation challenge.

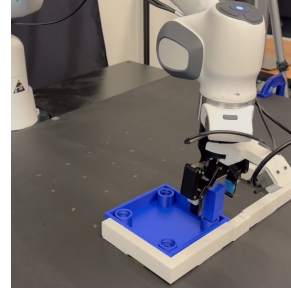


Figure 5: The real-world FurnitureBench square-table assembly task: the robot must grasp a table leg and screw it into a hole at the corner of the tabletop, a long-horizon contact-rich task that demonstrates DARP transfers directly to physical hardware.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: **DARP consistently outperforms BC across simulated and real-world manipulation.** Success rates (%) using low-dimensional state features across all three evaluation suites—RoboCasa, Robosuite, and a real-world FurnitureBench task (RoboCasa & Robosuite: 100 trials; Real: 50 trials); the higher success rate per task is in bold. DARP improves on BC in every one of the seven tasks and environments, and more than doubles its success rate on real hardware (44 → 92)—establishing that the gains persist under the sim-to-real gap. RoboCasa and Robosuite policies use Multi-Layer Perceptrons; the real-world results use a diffusion policy (see Section 8).

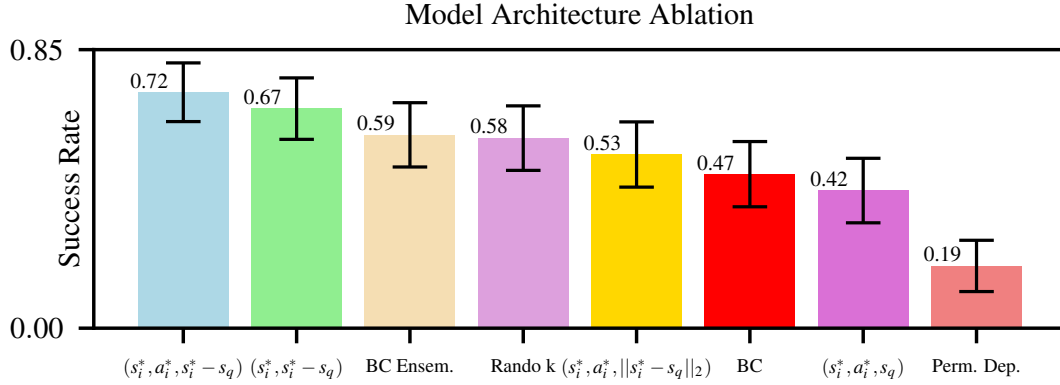


Figure 6: **Distance vectors and permutation invariance are the dominant contributors to DARP’s success.** Perturbing each core architectural choice in turn shows that DARP’s advantage is attributable primarily to conditioning on the difference-based tuple $(s_i^*, a_i^*, s_i^* - s_q)$: removing the distance vector or replacing the permutation-invariant aggregator sharply degrades performance, whereas removing the neighbor actions has only a modest effect. Success rate is averaged over 100 trials on the Robosuite Stack environment with 95% confidence intervals; higher is better.

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations. To establish that DARP applies well beyond the regime of compact, low-dimensional states, we evaluate it on simulated Robosuite manipulation using R3M image embeddings (Nair et al., 2022), which represent each state as a high-dimensional feature vector extracted from visual observations (Table 3). DARP not only outperforms standard BC on every vision-based task; its average improvement of $\sim 35\%$ actually *exceeds* the $\sim 22\%$ average improvement it achieves on the corresponding low-dimensional Robosuite tasks. The neighbor-based formulation therefore adapts to complex, high-dimensional state representations more effectively than standard BC does, and the effect is systematic rather than incidental: the setting in which BC is most brittle is precisely the setting in which DARP’s relative advantage is largest. This answers the first half of Q2 affirmatively—DARP scales to high-dimensional visual inputs and does so with a widening margin.

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: Success rates (%) on vision-based Robosuite tasks using R3M image embeddings; the higher value per task is in bold. DARP outperforms BC on all three tasks, and by a wider relative margin than in the low-dimensional setting.

Multi-modal Action Distributions. Equipping DARP with a parametric aggregator (Section 2.4) further enables it to model genuinely multimodal action distributions: on the multimodal Push-T environment, DARP exceeds behavior cloning by over 20%. This completes the answer to Q2—DARP scales along both axes, complex state representations and expressive action distributions alike. Full details appear in Appendix A.2.2.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: To identify which components of the DARP architecture are responsible for its gains, we conduct a comprehensive ablation that perturbs each design choice in isolation while holding the rest fixed: (1) standard DARP; (2) DARP without the neighbor actions; (3) an ensemble of 10 BC agents; (4) DARP with random neighbor selection in place of the distance metric; (5) DARP conditioned on the L_2 norm of the distance vector rather than the vector itself; (6) the BC baseline, which conditions only on the query state s_q ; (7) DARP conditioned on the query state instead of the difference vector between query and neighbor; and (8) DARP with a permutation-*dependent* aggregator in place of the permutation-invariant one. Because each variant removes exactly one ingredient, the resulting degradation measures that ingredient’s contribution directly; Figure 6 reports the results of this systematic ablation.

The ablation delivers a clear verdict on DARP’s mechanism and thereby answers Q3. Difference vectors and permutation invariance are the decisive ingredients, whereas the neighbor actions contribute more modestly. Random neighbor selection performs poorly, confirming that a meaningful distance metric—and not mere access to a neighborhood—is what renders retrieval informative. The permutation-invariant aggregator g is equally critical: permutation-*dependent* alternatives degrade performance substantially, corroborating the set-structured design motivated in Section 2.4. Both findings align precisely with the two design choices identified as central in the method, closing the loop between the architecture and its measured effect.

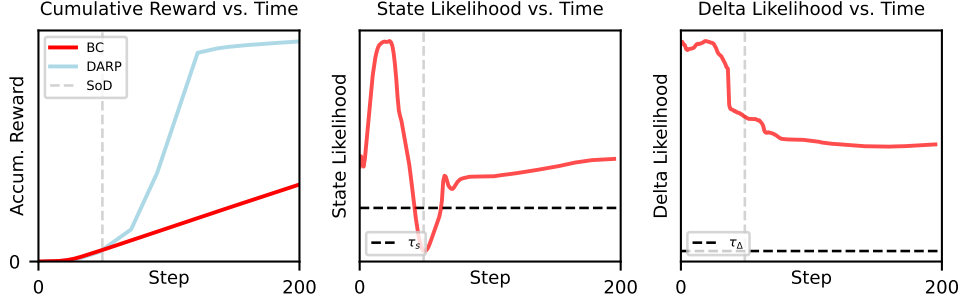


Figure 7: Cumulative rewards for BC and DARP on the Robosuite Stack task: initially identical rollouts diverge as BC fails and DARP succeeds. The vertical dashed line marks the “step of divergence” (SoD). At the SoD the query state’s likelihood falls below τ_s (out-of-distribution in absolute state coordinates), yet its difference-vector likelihood remains above τ_Δ (in-distribution once reparameterized)—exactly the regime in which DARP’s difference-based reparameterization is predicted to help, and in which it does.

Divergence Analysis: To pinpoint the mechanism behind DARP’s advantage over standard BC, we analyze the point of divergence in rollouts where BC fails but DARP succeeds. We define the “step of divergence” (SoD) as the step at which the two policies begin to accrue significantly different reward. Setting τ_s and τ_Δ to the 1st percentile of the training-set state and difference-vector likelihoods, respectively (so that 1% of training-time differences fall below τ_Δ), we uncover a strikingly consistent pattern: in all six rollouts across the two tasks (Robosuite Stack and MuJoCo Hopper), the query state at the SoD has state likelihood $< \tau_s$ yet difference-vector likelihood $> \tau_\Delta$. In other words, the failure states that defeat BC are out-of-distribution in absolute state coordinates but remain in-distribution once reparameterized as differences to retrieved neighbors. This is direct mechanistic evidence that DARP’s gains stem from improved prediction on the slightly out-of-distribution states enabled by the difference-based reparameterization—the very reparameterization that the ablation identifies as decisive (Figure 7 plots reward drift, the SoD, and the state and difference likelihoods for one task). Appendix A.2.4 reports additional experiments corroborating DARP’s robustness.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric IL algorithms achieve strong performance by exploiting local structure in the demonstration data. VINN Pari et al. (2022) applies locally weighted regression to imitation, obtaining surprisingly strong results in image embedding spaces, while MiDiGaP von Hartz et al. (2025) models multimodal trajectories with mixtures of Gaussian processes to enable rapid generalization. SEABO Lyu et al. (2024) repurposes retrieval for offline RL, constructing a reward function that rewards transitions lying close to their neighbors. FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) perform non-parametric retrieval and finetuning from large unlabeled datasets, generalizing through test-time training. DARP departs from all of these along two axes: it reparameterizes each retrieved neighbor as a difference-based tuple $(s_i^*, a_i^*, s_i^* - s_q)$, and it learns a genuinely semi-parametric policy rather than relying on purely non-parametric aggregation or test-time training. This combination is what secures the variance reduction characteristic of local methods together with the generalization of parametric policies—the two properties that prior non-parametric work obtains only individually.

Smoothness-Constrained Policy Learning: A substantial literature enforces explicit smoothness constraints to improve policy stability and robustness. L2C2 Kobayashi (2022) imposes local Lipschitz-continuity constraints in model-free RL, achieving smoothness and noise robustness without

sacrificing expressiveness, and Asadi et al. (2018) develop an analogous Lipschitz-constrained formulation for model-based RL. CCIL Ke et al. (2024a) extends these ideas to imitation learning, generating synthetic corrective labels via a Lipschitz-constrained dynamics model, and the approach has been scaled to humanoid controllers Chen et al. (2024) to reduce deployment-time shakiness. DARP differs fundamentally: rather than imposing an explicit constraint or auxiliary loss, it induces the same smoothness *implicitly* through an architectural change while training under the standard imitation learning objective, thereby dispensing with the additional hyperparameters these methods require. This implicit route not only avoids the extra machinery but matches or exceeds CCIL’s relative gains over BC across the standard MuJoCo tasks.

In-Context Learning Methods: A parallel line of work approaches retrieval through the lens of in-context imitation learning. REGENT Sridhar et al. (2025) studies retrieval-augmented generalization by feeding retrieved states, actions, and rewards into a causal transformer, and DPT Lee et al. (2023) employs supervised transformer pretraining to predict actions from a query state and an in-context dataset, effectively learning to explore. Related in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). Whereas these methods target rapid adaptation to new tasks and environments, DARP pursues a distinct objective—maximizing performance and stability within standard imitation learning—and its difference-based, permutation-invariant design follows directly from that focus. This distinction is borne out empirically: the in-context REGENT baseline trails DARP by a wide margin on the standard-IL setting for which DARP is designed.

5 CONCLUSION

We introduced **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a nearest-neighbor-based algorithm that reparameterizes imitation learning in terms of the relative differences between a query state and its retrieved neighbors rather than learning a single direct state-to-action mapping. We proved that this construction implicitly performs Laplacian smoothing—and thereby reduces variance and improves stability—while operating strictly within the assumptions of standard behavior cloning. Our evaluation across continuous control and robotic manipulation, spanning simulation and real hardware, substantiates three claims. First, DARP consistently outperforms standard behavior cloning under low-dimensional state representations. Second, this advantage not only persists but widens across high-dimensional visual representations, multimodal action distributions, and increasing task complexity, yielding improvements of 15–46% throughout the tested scenarios. Third, controlled ablations establish that the difference vectors and permutation-invariant aggregation are the components responsible for these gains, and a divergence analysis traces the mechanism to improved prediction on the slightly out-of-distribution states into which BC drifts. Together these results establish that reusing demonstration data through difference-aware retrieval is a general and practical route to more robust imitation learning—one that requires no additional supervision, no task-specific structure, and no smoothness hyperparameter, and that composes with modern generative policy classes rather than competing with them.

6 REPRODUCIBILITY STATEMENT

To ensure full reproducibility, we will release upon acceptance the complete supplementary source code, the policy and environment configuration files, the MuJoCo experiment data, the scripts that generate expert demonstrations for the Robosuite tasks via MimicGen, and the modality-transformation code. On NVIDIA L40 and L40s GPUs, the released artifacts reproduce the reported results exactly, the sole exception being the transformer-based methods (REGENT and the Set Transformer), whose non-deterministic execution may induce slight deviations from the reported numbers.

REFERENCES

- Kavosh Asadi, Dipendra Misra, and Michael Littman. Lipschitz continuity in model-based reinforcement learning. In *International Conference on Machine Learning*. PMLR, PMLR, 2018.
- Mikhail Belkin and Partha Niyogi. Towards a theoretical foundation for laplacian-based manifold methods. *Journal of Computer and System Sciences*, 74(8):1289–1308, 2008.

- Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0 : A vision-language-action flow model for general robot control, 2024. URL <https://arxiv.org/abs/2410.24164>.
- Zixuan Chen, Xialin He, Yen-Jen Wang, Qiayuan Liao, Yanjie Ze, Zhongyu Li, S. Shankar Sastry, Jiajun Wu, Koushil Sreenath, Saurabh Gupta, and Xue Bin Peng. Learning smooth humanoid locomotion through lipschitz-constrained policies. *arxiv preprint arXiv:2410.11825*, 2024.
- Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 2024.
- Fan R. K. Chung. *Spectral Graph Theory*, volume 92 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, 1997. doi: 10.1090/cbms/092.
- Michael Jae-Yoon Chung, Maxwell Forbes, Maya Cakmak, and Rajesh PN Rao. Accelerating imitation learning through crowdsourcing. In *2014 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4777–4784. IEEE, 2014.
- Norman Di Palo and Edward Johns. Keypoint action tokens enable in-context imitation learning in robotics. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024.
- Maximilian Du, Suraj Nair, Dorsa Sadigh, and Chelsea Finn. Behavior retrieval: Few-shot imitation learning by querying unlabeled datasets. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- Justin Fu, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine. D4rl: Datasets for deep data-driven reinforcement learning, 2020.
- Letian Fu, Huang Huang, Gaurav Datta, Lawrence Yunliang Chen, William Chung-Ho Panitch, Fangchen Liu, Hui Li, and Ken Goldberg. In-context imitation learning via next-token prediction. *arXiv preprint arXiv:2408.15980*, 2024.
- Liyiming Ke, Yunchu Zhang, Abhay Deshpande, Siddhartha Srinivasa, and Abhishek Gupta. Ccil: Continuity-based data augmentation for corrective imitation learning. In *The Twelfth International Conference on Learning Representations*, 2024a. URL <https://openreview.net/forum?id=LQ6LQ8f4y8>.
- Liyiming Ke, Yunchu Zhang, Abhay Deshpande, Siddhartha S. Srinivasa, and Abhishek Gupta. CCIL: continuity-based data augmentation for corrective imitation learning. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024b. URL <https://openreview.net/forum?id=LQ6LQ8f4y8>.
- Taisuke Kobayashi. L2c2: Locally lipschitz continuous constraint towards stable and smooth reinforcement learning. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4032–4039. IEEE, IEEE, October 2022.
- Jonathan Lee et al. Supervised pretraining can learn in-context reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pp. 43057–43083, 2023.
- Juho Lee, Yoonho Lee, Jungtaek Kim, Adam Kosior, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, pp. 3744–3753, 2019.
- Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. Offline reinforcement learning: Tutorial, review, and perspectives on open problems, 2020. URL <https://arxiv.org/abs/2005.01643>.

-
- Li-Heng Lin, Yuchen Cui, Amber Xie, Tianyu Hua, and Dorsa Sadigh. Flowretrieval: Flow-guided data retrieval for few-shot imitation learning. In Pulkrit Agrawal, Oliver Kroemer, and Wolfram Burgard (eds.), *Conference on Robot Learning, 6-9 November 2024, Munich, Germany*, volume 270 of *Proceedings of Machine Learning Research*, pp. 4084–4099. PMLR, 2024. URL <https://proceedings.mlr.press/v270/lin25a.html>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR) 2023*, 2023. URL <https://arxiv.org/abs/2210.02747>. Originally appeared as arXiv preprint arXiv:2210.02747.
- Jiafei Lyu, Xiaoteng Ma, Le Wan, Runze Liu, Xiu Li, , and Zongqing Lu. Seabo: A simple search-based method for offline imitation learning. In *International Conference on Learning Representations*, 2024.
- Ajay Mandlekar, Soroush Nasiriany, Bowen Wen, Iretiayo Akinola, Yashraj Narang, Linxi Fan, Yuke Zhu, and Dieter Fox. Mimicgen: A data generation system for scalable robot learning using human demonstrations. In *7th Annual Conference on Robot Learning*, 2023.
- Elman Mansimov and Kyunghyun Cho. Simple nearest neighbor policy method for continuous control tasks. In *International Conference on Learning Representations (ICLR) 2018*, 2018. Under review as a conference paper at ICLR 2018.
- Marius Memmel, Jacob Berg, Bingqing Chen, Abhishek Gupta, and Jonathan Francis. Strap: Robot sub-trajectory retrieval for augmented policy learning. In *The Thirteenth International Conference on Learning Representations*, 2025.
- Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3M: A universal visual representation for robot manipulation. In Karen Liu, Dana Kulic, and Jeffrey Ichnowski (eds.), *Conference on Robot Learning, CoRL 2022, 14-18 December 2022, Auckland, New Zealand*, volume 205 of *Proceedings of Machine Learning Research*, pp. 892–909. PMLR, 2022. URL <https://proceedings.mlr.press/v205/nair23a.html>.
- Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. Robocasa: Large-scale simulation of everyday tasks for generalist robots. In *Robotics: Science and Systems (RSS)*, 2024.
- Jyothish Pari, Nur Muhammad Shafuallah, Sridhar Pandian Arunachalam, and Lerrel Pinto. The surprising effectiveness of representation learning for visual imitation. In *Robotics: Science and Systems (RSS)*. Robotics: Science and Systems Foundation, June 2022. doi: 10.15607/RSS.2022.XVIII-052.
- Emmanuel Pignat and Sylvain Calinon. Bayesian gaussian mixture model for robotic policy imitation. *IEEE Robotics and Automation Letters*, 2019. URL <https://arxiv.org/pdf/1904.10716.pdf>. Preprint version.
- Dean Pomerleau. Efficient training of artificial neural networks for autonomous navigation. *Neural Comput.*, 3(1):88–97, 1991. doi: 10.1162/NECO.1991.3.1.88. URL <https://doi.org/10.1162/neco.1991.3.1.88>.
- Stéphane Ross and Drew Bagnell. Efficient reductions for imitation learning. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*, pp. 661–668. JMLR Workshop and Conference Proceedings, 2010.
- Stéphane Ross, Geoffrey J. Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In Geoffrey J. Gordon, David B. Dunson, and Miroslav Dudík (eds.), *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, AISTATS 2011, Fort Lauderdale, USA, April 11-13, 2011*, volume 15 of *JMLR Proceedings*, pp. 627–635. JMLR.org, 2011. URL <http://proceedings.mlr.press/v15/ross11a/ross11a.pdf>.

-
- Steven L. Salzberg and David W. Aha. Learning to catch: Applying nearest neighbor algorithms to dynamic control tasks. In P. Cheeseman and R. W. Oldford (eds.), *Selecting Models from Data: Artificial Intelligence and Statistics IV*, volume 89 of *Lecture Notes in Statistics*, pp. 321–328. Springer, 1994. doi: 10.1007/978-1-4612-2660-4_33. URL https://link.springer.com/chapter/10.1007/978-1-4612-2660-4_33.
- Jonathan C. Spencer, Sanjiban Choudhury, Arun Venkatraman, Brian D. Ziebart, and J. Andrew Bagnell. Feedback in imitation learning: The three regimes of covariate shift. *CoRR*, abs/2102.02872, 2021. URL <https://arxiv.org/abs/2102.02872>.
- Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, and Insup Lee. REGENT: A retrieval-augmented generalist agent that can act in-context in new environments. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. Oral presentation.
- Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ international conference on intelligent robots and systems*, pp. 5026–5033. IEEE, 2012.
- Arun Venkatraman, Martial Hebert, and J. Andrew Bagnell. Improving multi-step prediction of learned time series models. In Blai Bonet and Sven Koenig (eds.), *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence, January 25-30, 2015, Austin, Texas, USA*, pp. 3024–3030. AAAI Press, 2015. doi: 10.1609/AAAI.V29I1.9590. URL <https://doi.org/10.1609/aaai.v29i1.9590>.
- Jan Ole von Hartz, Adrian Röfer, Joschka Boedecker, and Abhinav Valada. The unreasonable effectiveness of discrete-time gaussian process mixtures for robot policy learning. *arXiv preprint arXiv:2505.03296*, 2025.
- Vitalis Vosylius and Edward Johns. Instant policy: In-context imitation learning via graph diffusion. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan R Salakhutdinov, and Alexander J Smola. Deep sets. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett (eds.), *Advances in Neural Information Processing Systems 30*, pp. 3391–3401. Curran Associates, Inc., 2017. URL <http://papers.nips.cc/paper/6931-deep-sets.pdf>.
- Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- Dengyong Zhou, Olivier Bousquet, Thomas Navin Lal, Jason Weston, and Bernhard Schölkopf. Learning with local and global consistency. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 16, 2003.
- Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Soroush Nasiriany, Yifeng Zhu, and Kevin Lin. robosuite: A modular simulation framework and benchmark for robot learning. In *arXiv preprint arXiv:2009.12293*, 2020.

A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

The proof of Theorem 1 rests on a well-established result (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008), which we state first and then invoke.

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2} (\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \longrightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.

(iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Taken together, claims (i)–(iii) establish that MRIL enjoys strictly stronger generalization and stability guarantees than vanilla BC.

Proof. We prove each claim in turn.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top L f,$$

where $L = D - W$ is the graph Laplacian, and W is the k -NN affinity matrix. By Lemma 1, this equals the empirical Dirichlet energy of f on the k -NN graph. It is well known (Zhou et al., 2003; Belkin & Niyogi, 2008) that such a quadratic penalty is equivalent to Tikhonov regularization with respect to the graph Laplacian norm $\|f\|_L^2 = f^\top L f$. In statistical learning theory, adding a Tikhonov penalty strictly reduces the variance of the estimator compared to the unregularized solution while keeping the bias term controlled. Thus MRIL enjoys smaller estimator variance than vanilla BC, which uses no such penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

This is the weighted Dirichlet energy of f on $\mathcal{M}$. If this integral is finite, f belongs to the Sobolev space $H^1(\mathcal{M}, p^2)$, and in particular f is locally Lipschitz almost everywhere with

$$\|f(s^*) - f(s'^*)\| \leq C \|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Therefore minimizers of $\mathcal{L}_{\text{MRIL}}$ have uniformly bounded local Lipschitz constants along high-density regions of the state space. By contrast, minimizers of vanilla BC have no such constraint: any oscillatory interpolant that matches the training data exactly yields the same supervised risk, so arbitrarily large Lipschitz constants are possible.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . For Lipschitz dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is only minimized on the empirical distribution $P_{\mathcal{S}}$; off-distribution, it may be $O(1)$ regardless of Δ_t . The second term satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Hence the recursion can take the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a(O(h^2) + O(\Delta_t)).$$

Since the constant multiplying Δ_t is strictly smaller under the smoothness constraint, the cumulative error grows strictly slower than in vanilla BC. In particular, error growth is sublinear in the rollout horizon when h is small, whereas it is linear for vanilla BC.

Conclusion. Claims (i)–(iii) establish that MRIL yields lower variance, uniform smoothness control, and sublinear rollout error accumulation compared to vanilla BC, completing the proof. $\square$

Kernel choice. For the smoothness regularizer we adopt a Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i^*\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h fixed to the median distance to the k -th nearest neighbor over the dataset. This choice is standard in manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and guarantees convergence of the graph Laplacian penalty to the Dirichlet energy in the continuum limit. Empirically, this default proved stable across every task, and alternative kernels (e.g., uniform k -NN or exponential decay) yielded qualitatively identical behavior, indicating that the effect is a property of the neighborhood structure rather than of the specific kernel.

A.1.2 PROOF OF THEOREM 2

We first establish the two lemmas that characterize the spectral form of, respectively, explicit Laplacian regularization and neighbor aggregation; the equivalence in Theorem 2 then follows by comparing the two filters.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly determines the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U\Lambda U^\top$ with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. Write $f = Uc$, $a^* = Ub$ in this basis. The objective becomes

$$\|Uc - Ub\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$. Transforming back gives the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = Sf$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2}AD^{-1/2}$ and $S = D^{-1}A = I - L_{\text{rw}}$ where $L_{\text{rw}} = D^{-1}L$ is the random-walk Laplacian. Since L_{rw} and L share the same spectrum up to similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Thus for each mode u_j , $Su_j = (1 - \mu_j)u_j$, yielding the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Thus in the graph Fourier basis, neighbor aggregation corresponds to multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

On the other hand, Lemma 2 shows that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight high-frequency modes ($\mu \gg 0$) while preserving low-frequency modes ($\mu \approx 0$). The key difference is that $\phi_\lambda(\mu)$ requires tuning λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP can be interpreted as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. Moreover, for large μ , $\phi_{\text{DARP}}(\mu)$ damps high-frequency modes even more strongly by driving them toward zero, providing a sharper low-pass effect than explicit regularization.

Therefore, DARP's aggregation step is mathematically equivalent to implicit Laplacian regularization with fixed spectral filter ϕ_{DARP} , eliminating the need to tune λ explicitly. $\square$

DARP is therefore best understood as a form of *locally adaptive implicit regularization*: in place of an explicit global weight λ , its neighbor aggregation step enforces smoothness automatically through the graph structure, with an effective regularization strength that varies with the local degree and neighborhood geometry and thereby adapts to the density of the expert demonstrations. Spectrally, this reduces to the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$, which suppresses high-frequency modes more aggressively than any fixed explicit λ . Figure 8 renders this comparison concrete, showing that DARP attains sharper low-pass filtering with no hyperparameter tuning whatsoever—the practical advantage that motivates the implicit construction.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

The distance function $d(s_q, s_i^*)$ used to select the k neighbors is a crucial design decision. Although a plain Euclidean distance between individual states is already effective, our experiments employ a slightly refined procedure that exploits the sequential structure of the data by incorporating trajectory

history into the distance computation.

Consider a query trajectory $S_q = (\dots, s_{q,-1}, s_{q,0})$ in which $s_{q,0}$ is the current query state s_q . To evaluate $d(s_q, s_i^*)$ for an expert state s_i^* , we first identify the source trajectory S_j^* containing s_i^* together with its index t within that trajectory, so that s_i^* may be written as $s_{j,t}^*$. Given a lookback parameter ℓ that specifies how many past states to consider, the history-augmented distance is

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

which is simply the accumulated Euclidean distance between the current and preceding $\ell - 1$ states of the query and source trajectories, taken over valid indices. Because more recent states should exert greater influence on neighbor selection, we additionally down-weight older states with an exponential decay rate $\gamma \geq 0$, giving

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

Figure 9 reports how environment success rate varies as each of these retrieval parameters is swept in turn.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

We probe DARP’s capacity to represent complex action distributions on the Push-T task of (Chi et al., 2024), which is inherently multimodal (Figure 10). Here DARP is equipped with a Set Transformer head that predicts the parameters of a Gaussian Mixture Model. This configuration handles the multimodal distribution effectively, improving over BC by 22% (Q2; Table 4) and thereby confirming that the aggregation operator generalizes cleanly from unimodal regression to expressive multimodal action modeling—precisely the extension anticipated by the parametric-aggregator formulation.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A natural concern for any neighbor-based method is its behavior in environments with strong discontinuities, where states that are close in Euclidean distance may nonetheless require drastically different actions. We construct a deliberate stress test for exactly this failure mode using a modified version of D4RL’s Umaze environment (Figure 11), designed to be maximally adversarial to retrieval.

Even in this adversarially discontinuous environment, DARP attains a 57% success rate against BC’s 25%, more than doubling BC’s performance (Q3; Table 5). The result shows that the difference vectors and permutation-invariant aggregation together enable the model to discriminate appropriate from inappropriate neighbors even when spatial proximity does not imply action similarity—the very regime that most challenges naive retrieval, and in which purely non-parametric neighbor methods break down.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

To isolate DARP’s robustness to accumulated error, we roll out a BC agent in an environment where it is known to fail and, every k steps, fork the environment and launch a DARP agent from the cloned state. The results (Fig. 12) show that even as BC drifts away from the support of the expert demonstrations toward failure, DARP recovers from those same states and scores highly. This directly substantiates the central claim that DARP is markedly more robust to error accumulation, performing reliably in precisely the slightly out-of-distribution states into which a failing BC agent drifts—the same regime pinpointed by the divergence analysis in the main text.

A.2.5 COMPARISON WITH CCIL

Table 6 compares DARP against CCIL, a strong baseline that induces smoothness explicitly through a Lipschitz-constrained dynamics model. Using the scores reported in the CCIL paper and comparing percentage improvement over BC, DARP outperforms CCIL on three of the four environments and

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC: DARP versus CCIL.** Comparison on standard MuJoCo benchmarks, reported as percentage improvement over BC (best per environment in bold). DARP matches or exceeds CCIL’s relative gain on every task, with an especially decisive margin on HalfCheetah (418.7% vs. 5.4%).

matches it on the fourth (Walker), with a particularly large margin on HalfCheetah (418.7% vs. 5.4%). DARP’s *implicit* smoothing therefore not only dispenses with the extra machinery CCIL requires but also delivers strictly larger relative gains, directly validating the implicit-over-explicit design at the heart of the method.

A.2.6 DISTANCE METRIC SENSITIVITY

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance Metric Sensitivity.** Success rates on the Robosuite Stack task using R3M features, best in bold. DARP performs comparably under Euclidean distance and cosine similarity, exceeding BC by 28% and 23% respectively—evidence of robustness to the choice of retrieval metric.

Although every experiment in this paper selects neighbors by Euclidean distance, alternative metrics such as cosine similarity are natural to consider, particularly in high-dimensional embedding spaces like R3M. Table 7 shows that DARP performs comparably under the two metrics—cosine similarity trails Euclidean distance by only 5% while still exceeding BC by 23%—establishing that DARP’s advantage is a robust property of the method rather than an artifact of a particular distance metric.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP composes with diffusion policy.** Success rates (%) over 50 trials on the state-based Robosuite Stack, Threading, and Peg Insertion tasks, best per task in bold. DARP improves both a standard MLP backbone and a diffusion-policy backbone, demonstrating that its difference-aware retrieval is complementary to—not competitive with—state-of-the-art generative policy classes. The demonstration count here is substantially smaller than in Table 2, so the BC and DARP numbers are not directly comparable across the two tables.

Although our main experiments use an MLP backbone, diffusion policy (Chi et al., 2024) has emerged as a state-of-the-art model class for imitation learning, especially for manipulation. Table 8 shows that DARP is fully compatible with diffusion and, in combination, yields an even stronger model: the DARP architecture with a diffusion backbone improves over DARP with an MLP backbone by 20%, 34%, and 26% success rate on the three tasks, and over standard BC by 48%, 60%, and 38%, respectively. DARP therefore acts as a general enhancement that stacks with modern generative policy classes rather than replacing them—an orthogonality that broadens its applicability across the current imitation-learning toolkit.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational efficiency: DARP (MLP) versus diffusion policy.** Per-epoch training and per-step inference runtimes, best in bold. DARP with an MLP backbone is dramatically cheaper— $\approx 10\times$ faster to train and $\approx 36\times$ faster at inference—so its accuracy gains impose no cost in computational tractability.

Moreover, DARP with an MLP backbone is substantially faster than standard diffusion, most notably at inference time (Table 9), which makes it directly suitable for real-time robotic control where a diffusion backbone would be prohibitive.

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results with human demonstrations.** Success rates on the Robosuite Stack task when training on human-collected data rather than RL-generated data, best in bold. DARP retains a 15% advantage over BC, confirming that its gains do not depend on the demonstrations being machine-generated.

The expert demonstrations for our MuJoCo and Robosuite experiments are generated by an optimal reinforcement learning policy. Because human demonstrations differ systematically in optimality and consistency, it is important to verify that DARP’s advantage survives this shift in data source. It does: trained on human demonstrations for the Robosuite Stack task, DARP outperforms standard BC by 15% (Table 10), confirming that the method’s benefits extend to realistic, human-provided data and are not tied to the near-optimality of RL-generated experts.

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Validation-loss-based selection of training epochs and retrieval hyperparameters.** Mean scores on the Hopper and Stack environments, best in bold. DARP retains its advantage over BC under this principled, held-out selection criterion, establishing that its gains do not rely on oracle hyperparameter choices.

Figure 9 shows that extreme retrieval settings—for instance, a very small number of neighbors—can drive DARP below standard BC. This dependence poses no practical obstacle: selecting the retrieval

hyperparameters that minimize validation loss reliably identifies performant settings, and under this held-out criterion DARP retains its advantage over BC (Table 11). The gains thus reflect the method itself, not privileged access to oracle hyperparameters.

A.3 PSEUDOCODE

Algorithm 1 specifies the complete training procedure for DARP, covering both the fixed-average and parametric-aggregator instantiations within a single unified loop.

Algorithm 1 Difference-Aware Retrieval Policies

```

1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \arg \min -k, d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

```

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime scaling with neighbor count k .** Per-epoch training and per-step inference times (seconds) for BC and for DARP across a range of k on the Hopper and Stack datasets. Cost grows sub-linearly in k , and even at large k inference remains well within real-time control budgets.

Table 12 quantifies the computational overhead of DARP at both training and inference time. The cost scales sub-linearly in the neighbor count k , and inference remains comfortably within the budget for real-time robotic control: on the Robosuite Stack task at $k = 500$, a single inference step takes approximately 0.00437 seconds on our hardware, corresponding to a control frequency exceeding 230 Hz. DARP therefore attains its accuracy and robustness gains at negligible computational cost, preserving full deployability on real-time control loops.

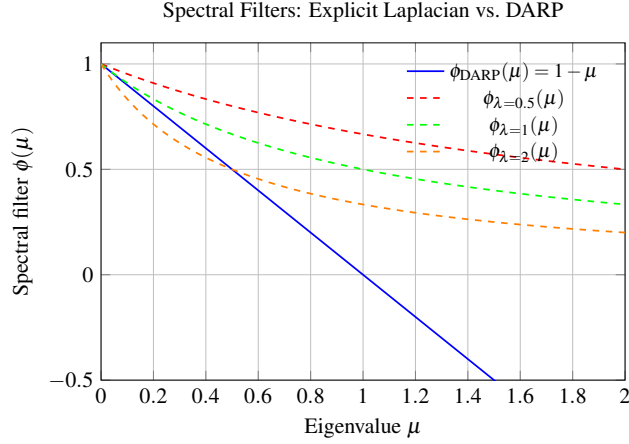


Figure 8: Spectral filters of DARP’s fixed aggregation ($\phi_{\text{DARP}}(\mu) = 1 - \mu$) versus explicit Laplacian regularization at several values of λ : DARP achieves sharper low-pass filtering, damping high-frequency modes more aggressively while requiring no hyperparameter tuning whatsoever.

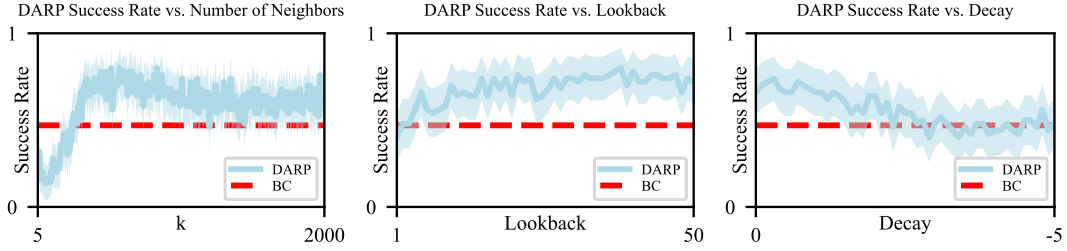


Figure 9: DARP performance as each of the three retrieval hyperparameters is swept, on the Robosuite Stack environment (success rate over 50 trials, 95% confidence intervals; higher is better). **(Left)** As a function of the neighbor count k , performance is poor for very few neighbors, attains a global optimum near $k \approx 500$, and plateaus just above BC’s success rate as k approaches the dataset size. **(Center)** As a function of the lookback horizon, performance improves modestly as more history is incorporated, and drops below BC only when little or no history is used. **(Right)** As a function of the history decay rate, performance is sensitive to the weight placed on older observations: an overly aggressive decay effectively removes the lookback, again pushing DARP below BC. Crucially, across the interior of every sweep DARP remains well above BC, so its advantage does not hinge on precise hyperparameter tuning.

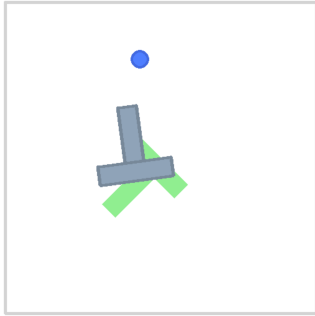


Figure 10: **Push-T Environment.** The agent controls the blue circle to push the T-shaped block into a target configuration—a task that demands genuinely multi-modal action prediction, since multiple push directions are valid from a given state.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T Results.** Score averaged over 100 trials; DARP (with a Set Transformer GMM head) outperforms BC by 22%, with the higher score in bold.

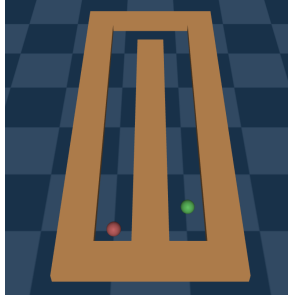


Figure 11: **Long maze environment.** A force-actuated ball must be driven from the green start to the red destination; the elongated corridors induce the state discontinuities—nearby coordinates on opposite walls demanding opposing actions—that stress-test neighbor-based prediction.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Success rate over 100 trials, best in bold; DARP more than doubles BC’s success rate (57% vs. 25%) even in this discontinuity-dominated setting.

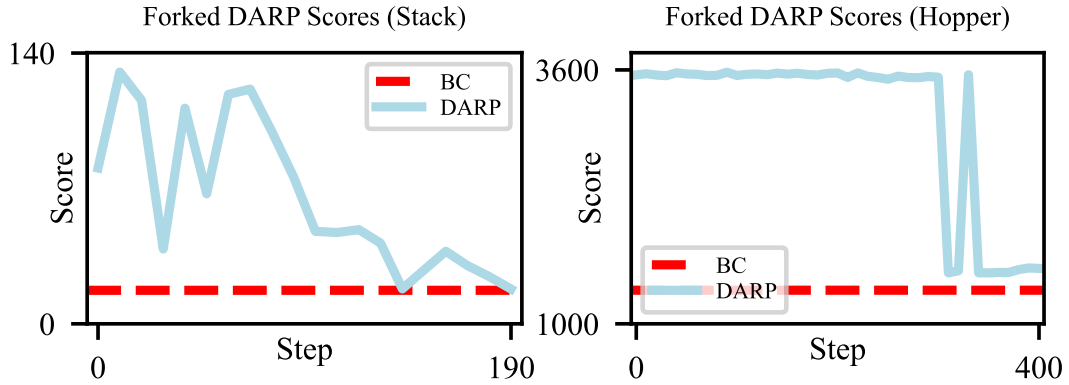


Figure 12: Recovery from BC-induced drift on two tasks (Robosuite Stack and MuJoCo Hopper). We roll out a BC agent and fork the environment every $k = 10$ steps, launching a DARP agent from each fork. Even as the BC rollout approaches failure, DARP recovers and scores highly from the forked states, forfeiting this ability only about halfway through the Stack rollout and about 80% of the way through the Hopper rollout—a direct quantification of DARP’s robustness to the accumulated error that defeats BC.